

Scalability & Future Plan

Date	15 July 2026
Team ID	XXXXXX
Project Name	AI Debt Settlement System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses SQLite database, which is suitable only for small to medium-scale applications.	Limited scalability and concurrent user support.	High
2	AI recommendations depend on an external Google Gemini API service.	API rate limits and internet dependency may affect response time.	Medium
3	Supports only basic user authentication without advanced security features like Multi-Factor Authentication (MFA).	Increased security risks for production environments.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of concurrent users using a single FastAPI server.	Deploy using Docker and Kubernetes with Load Balancers (Nginx) to support thousands of concurrent users.
2	Data Storage	Uses SQLite for storing user and loan information.	Migrate to PostgreSQL or MySQL with cloud-managed databases (AWS RDS/Azure SQL).
3	Performance	Backend processes requests sequentially with limited caching.	Implement Redis caching, asynchronous processing, and API optimization for faster response times.
4	Security	JWT authentication with password hashing	Introduce Multi-Factor Authentication (MFA), Role-Based Access Control (RBAC), HTTPS, and refresh

		using bcrypt.	tokens for enhanced security.
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the application on AWS/Azure with a production-ready PostgreSQL database.	1–2 Months	Improved reliability, scalability, and accessibility.
Phase 2	Introduce Role-Based Access Control (Admin, Financial Advisor, User).	2–3 Months	Better security and efficient management of multiple user roles.
Phase 3	Integrate Bank APIs and Credit Bureau APIs for automatic loan synchronization and credit score analysis.	3–5 Months	Eliminates manual data entry and provides real-time financial insights.
Phase 4	Develop Android and iOS mobile applications using React Native or Flutter.	5–6 Months	Increased accessibility and improved user experience across mobile devices.
Phase 5	Enhance AI capabilities using predictive analytics, repayment forecasting, and personalized financial planning.	6–9 Months	Smarter recommendations, improved financial decision-making, and better debt management outcomes.